

ANALYSTLAB AFRICA

Data Science Internship Program

Week 5 Project Summary

HealthConnect Clinic — Data Science Track

What I Planned to Accomplish in Week 5

Move from Week 4's problem definition to practical baseline model development: formal data preparation, feature engineering, a defined train/test strategy, at least one baseline classification model, and initial evaluation using standard classification metrics.

What I Actually Completed

- Reviewed and confirmed the Week 4 problem definition, target variable, and feature list
- Cleaned the dataset by applying structural imputation for reminder_channel, median imputation for distance_to_clinic_km, correcting date types, and removing identifiers and the confirmed leakage field (waiting_time_minutes)
- Produced EDA with interpreted visualizations, including distributions, boxplots, categorical/target relationships, and a correlation heatmap
- Engineered three features: is_new_patient, prior_no_show_rate, and lead_time_bucket
- Defined and justified a stratified random 80/20 train/test split
- Built and evaluated two baseline models: Logistic Regression and Random Forest
- Tested an alternative target-handling approach by merging Cancelled into No-Show) as a documented comparison, rather than a silent change

Key Findings and Development Outcomes

Baseline performance: Logistic Regression — Accuracy 0.616, Precision 0.614, Recall 0.672, F1 0.642, ROC-AUC 0.669

Model comparison: Logistic Regression outperformed Random Forest on every metric (RF ROC-AUC 0.635), suggesting the relationship is largely linear and additive rather than driven by complex interactions

Top predictors: booking_lead_days and previous_no_shows emerged as the strongest positive predictors of no-show risk, consistent with Week 4's informal signal checks.

Target-handling test: A merged Cancelled + No-Show target performed marginally worse (ROC-AUC 0.657 vs 0.669), and behavioral data showed that Cancelled patients resemble Attended patients more than No-Show patients

Major Challenges Encountered

- Deciding how to handle new patients (4.8% of the dataset) when engineering a no-show rate feature, since the ratio is mathematically undefined with zero prior appointments
- Confirming whether the previously flagged waiting_time_minutes leakage risk was truly safe to exclude, rather than assuming the Week 4 flag without re-verification

Important Decisions Made and Why

- Assigned new patients the dataset-wide average no-show rate, rather than 0 or a missing value, when engineering `prior_no_show_rate` to avoid implying guaranteed attendance for patients with no history
- Chose a stratified random split over a time-based split, based on evidence that no-show rates showed no meaningful drift across the 18-month date range in the data
- Retained Logistic Regression as the lead model for Week 6, based on its stronger and more interpretable baseline performance relative to Random Forest

Changes to the Week 4 Approach

No changes were made to the Week 4 target variable decision. The proposal to exclude Cancelled appointments was tested against an alternative (merging Cancelled into No-Show) and confirmed with new evidence: Cancelled patients showed a lower average prior no-show rate (0.418) than Attended patients (0.457), while true No-Show patients averaged 0.641, supporting the original decision rather than reversing it. All other Week 4 feature and modeling proposals were carried forward unchanged.

Cross-Track Collaboration Completed

Coordinated with the Data Analytics track: their Week 4 focus on no-show patterns by reminder status and distance to clinic informed the decision to retain `reminder_sent`, `reminder_channel`, and `distance_to_clinic_km` as candidate features in this notebook, despite their comparatively mild individual effect sizes in isolation.

Remaining Work

- Formal feature importance analysis to prioritize which engineered features earn their place in a refined model
- Hyperparameter tuning for the Logistic Regression baseline
- Testing a time-based train/test split for a more production realistic evaluation

Proposed Focus for Week 6

Refine the Logistic Regression baseline through feature importance analysis and hyperparameter tuning, evaluate a time-based train/test split for production realism, and coordinate with the Machine Learning Engineering track on the expected model output format before any integration work.